MIL-C-10479B(EL)

24 FEBRUARY 1969

SUPERSEDING
MIL-C-10479A(SigC)
15 August 1961

MILITARY SPECIFICATION

CRYSTAL IMPEDANCE METER TS-683( )/TSM

1. SCOPE.

1.1 This specification covers one type of test set used to measure crystal units in the frequency range from 10 MHz to 140 MHz and is designated Crystal Impedance Meter TS-683( )/TSM.

2. APPLICABLE DOCUMENTS.

2.1 Documents. The following documents of the issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping.
PPP-B-636	Box, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-S-760	Strapping, Nonmetallic, (and Connectors).
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 6625

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MIL-E-75	Electron, Tube Packing, Packaging, and Container Marking for General Specification for.
MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags and Sleeves, Interior Packaging.
MIL-C-3098	Crystal Units, Quartz.
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-B-43291	Boxes; Fiberboard; Corrugated, Double-Wall, Weather Resistant.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads; 40 in. x 48 in. 4-Way Partial and 4-Way Pallets.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-781	Reliability Tests Exponential Distribution.
MIL-STD-810	Military Standard Environmental Test Methods for Aerospace and Ground Equipment.

DRAWINGS

ELECTRONIC COMMAND

SC-A-46439	List of Accessories for Package Tester.
SC-DL-97713	Crystal Impedance Meter TS-683( )/TSM.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS.

3.1 Description. The Crystal Impedance Meter TS-683( )/TSM is used to measure the equivalent electrical parameters of quartz crystals of the type used for communication purposes. The equipment is designed specifically to test quartz crystal units for conformance with military specification. Provision is made to measure directly the effective series-resonant and antiresonant resistance of a piezoelectric quartz crystal. The equipment is designed to operate from 115 or 230 Vac, 50 to 400 Hz. The accessories consist of an rf output cable, one (1) crystal socket adapter UG-683( )/U, one (1) crystal socket adapter MX-3020( )/TSM, and a case that contains nine (9) calibrating resistors, one (1) antiresonant adapter, and one (1) variable calibrating resistor.

3.2 First article samples. The contractor shall furnish three (3) complete first article samples for approval provided the invitation for bid and the contract requires first article samples.

3.3 Construction. The equipment shall be constructed in accordance with SC-DL-97713 and the requirements specified herein.

3.3.1 Model. A model shall be supplied to the contractor as required to meet requirements of 3.7.3.

3.4 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements therein, shall apply.

3.5 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware; such as screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be final Film E, semigloss, light gray enamel, conforming to MIL-F-14072.

3.6 Marking. Marking shall conform to MIL-M-13231.

3.6.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.6.2 Serial numbers. Serial numbers are required on each nomenclatured and type designated item, except cords and cables.

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3.7 Electrical.

3.7.1 Frequency range. The crystal impedance meter shall cover a frequency range of 10 to 140 MHz in four bands as listed below. The accuracy of the generator shall not deviate by more than 10 percent as indicated by the frequency on an electronic counter when tested in accordance with 4.7.1.

Band	Calibrated dial frequency	Frequency coverage requirements
1	10 - 20 MHz	9.8 - 20 MHz
2	18 - 60 MHz	18 - 60 MHz
3	55 - 75 MHz	55 - 75 MHz
4	65 - 140 MHz	65 - 142.8 MHz

3.7.2 Amplitude of oscillation. The crystal impedance meter shall oscillate with an amplitude that shall cause a grid current reading of at least 20 percent of full scale deflection on the grid current meter with a resistance inserted into the crystal socket that corresponds to the particular frequency being used as listed below. The amplitude of oscillation measured shall be tested in accordance with 4.7.2.

Band	Frequency of crystal impedance meter	Calibrating resistor
1	10 - 20 MHz	100 ohms
2	18 - 60 MHz	100 ohms
3	55 - 65 MHz	100 ohms
3	65 - 75 MHz	82 ohms
4	60 - 100 MHz	100 ohms
4	100 - 140 MHz	40 ohms

3.7.3 Resistance determination.

3.7.3.1 Drive level. The drive level shall be set to within + 10.0 percent of the specified drive level when tested in accordance with 4.7.3.1.

3.7.3.2 Equivalent resistance of series resonant and anti-resonant crystals. The crystal impedance meter shall be capable of determining the equivalent resistance of series resonant and anti-resonant crystals in the frequency range of 10 MHz to 140 MHz. The difference between the reading obtained on the crystal impedance meter under test and the reading obtained on the government furnished Crystal Impedance Meter TS-683( )/TSM shall not deviate by more than ± 10 percent when tested in accordance with 4.7.3.2.

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3.7.4 Self oscillation. Uncontrolled oscillation shall not be present when there is no crystal inserted in the crystal socket and when tested in accordance with 4.7.4.

3.7.5 Calibration resistors. Nine calibration resistors are provided with each Crystal Impedance Meter TS-683( )/TSM. These resistors shall be mounted on bases which can be plugged into the crystal socket. Their values are 10, 22, 30, 40, 51, 60, 68, 82, and 100 ohms. They shall not deviate from their dc value when measured at 110 MHz by more than ±5 percent and shall have a shunt capacitance no greater than 0.6 pF when tested in accordance with 4.7.5. The dc value of these resistors shall not deviate by more than ±5 percent except the 40 ohm and 60 ohm resistors which shall not deviate by more than ±2 percent when tested in accordance with 4.7.5.

3.7.6 RF pickup cord assembly.

3.7.6.1 Pull strength. The terminated end of the cable assembly shall withstand a pull of 25 pounds for a period of 1 minute without displacing the center pin of the connector by more than 0.015 inch when tested in accordance with 4.7.6.1.

3.7.6.2 Dielectric strength. The dielectric between the inner and outer conductors shall be capable of withstanding a potential of 750 volts (rms) 60 Hz for a period of 1 minute without arcing or other indication of breakdown when tested in accordance with 4.7.6.2.

3.7.6.3 Insulation resistance. The insulation resistance shall not be less than 1000 megohms when tested in accordance with 4.7.6.3.

3.7.7 Power supply. The equipment shall operate from a 115/230 V ±10 percent, 50/400 Hz single phase source when tested in accordance with 4.7.7.

3.7.8 Calibration adaptors. Two adaptors are provided with each Crystal Impedance Meter TS-683( )/TSM.

(a) An adaptor consisting of a 100 ohm potentiometer mounted in a case and shall be capable of being inserted into the crystal socket for resonant measurements.

(b) An adaptor to be used for anti-resonant measurements, consisting of a crystal socket and a 32 pF capacitor with an accuracy of ±0.2 pF mounted in a case and so arranged that the 32 pF capacitor is in series with the crystal socket of the adaptor when the adaptor is inserted in the crystal socket of the impedance meter.

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(c) The adaptors shall be tested in accordance with 4.7.8.

3.8 Service conditions. The Crystal Impedance Meter TS-683( )/TSM shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified herein when subjected to the following service conditions.

3.8.1 Fungus. The equipment, including accessories, shall show no evidence of viable fungus and shall meet the requirements of 3.7.2 when tested in accordance with 4.8.1.

3.8.2 Low temperature. The equipment including accessories shall be subjected to the low temperature extremes of -4°F operating and -65°F storage and shall meet the electrical requirements of 3.7.2 at times specified when tested in accordance with 4.8.2.

3.8.3 High temperature. The equipment including accessories shall be subjected to the high temperature extremes of 125°F operating and 160°F storage and shall meet the electrical requirements of 3.7.2 at times specified when tested in accordance with 4.8.3.

3.8.4 Humidity. The equipment including accessories shall exhibit no physical damage of parts and material, such as corrosion, rust, blistering, swelling, deterioration, etc; or electrical degradation, and shall meet the requirements of 3.7.2 at times specified, when tested in accordance with 4.8.4.

3.8.5 Altitude. The equipment including accessories shall be subjected to the altitudes specified and shall meet the electrical requirements of 3.7.2 at times specified when tested in accordance with 4.8.5.

3.8.6 Shock, bench-handling. The equipment, less accessories, shall show no evidence of physical damage except minor surface abrasion and shall meet the requirements of 3.7.2 when tested in accordance with 4.8.6.

3.8.7 Vibration. The equipment, including accessories, shall show no evidence of breakage or loosening of parts, and shall meet the requirements of 3.7.2 when tested in accordance with 4.8.7.

3.8.8 Electromagnetic interference. The equipment shall comply with CE03, CS01, CS02, CS06, RE02-(.014-1000 MHz) and RS03 (.014-1000 MHz) interference and susceptibility requirements for Class 1C equipments of MIL-STD-461 when tested in accordance with 4.8.8.

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3.9 Visual and mechanical. The complete equipment shall meet the requirements of MIL-STD-252 as specified in 4.9.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (see 4.10).

3.11 Burn-in and debugging. Each equipment less accessories produced shall be subjected to the burn-in and debugging test of 4.11 after bounce preconditioning and prior to Group A inspection. The equipment shall be monitored for amplitude of oscillation per 3.7.2 and 4.7.2, during period of test.

3.12 Bounce preconditioning. Each equipment, including accessories, produced shall be subjected to the bounce preconditioning test of 4.6 prior to Group A inspection. After bounce preconditioning, the equipment, including accessories, shall not be repaired, aligned, cleaned or otherwise changed prior to performing burn-in and debugging, and Group A inspection.

3.13 Workmanship. The complete equipment shall be manufactured and assembled in accordance with the applicable portions of MIL-P-11268 as follows:

General requirements for plastic material and parts.
Wiring and cabling including:
- Slack.
- Protection.
- Clearance.
- Splicing.
- Connectors, general.
- Grounding, general.

Flux and cleaning agents for soldering.
Process for soldering.
Cleaning of equipment.
General process for securing of parts.
Process of welding.
Connectors, general.
Mechanical operation of controls.
Tropicalization of material.
Lubricants.

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4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the contract and specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (Does not include preparation for delivery) (see 4.3).

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables I, II and III respectively) on the three (3) samples of the completely assembled equipment, including accessories. The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government production engineer. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.6
Workmanship	3.13

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4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit produced shall be inspected by the contractor for conformance to all the Group A requirements and tests in Table I. Each lot shall be subjected to Government verification and this inspection, including sampling, shall conform to Table I and the ordinary inspection procedures of MIL-STD-105.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE I. Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL	
			Major	Minor
Visual and mechanical	3.9	4.9	1 dphu*	4 dphu
Electrical				
Frequency range	3.7.1	4.7.1		
Amplitude of oscillation	3.7.2	4.7.2	1 dphu for the group	**
Resistance determination	3.7.3	4.7.3		
Self-oscillation	3.7.4	4.7.4		
Power supply	3.7.7	4.7.7		

* Defects per hundred units.
** All electrical tests are major.

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and meet the Group A inspection.

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4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE II. Group B inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability	3.10	4.10	4 dphu*
Electrical			
Calibration resistors	3.7.5	4.7.5	
RF pickup cord assembly			
Pull strength	3.7.6.1	4.7.6.1	
Dielectric strength	3.7.6.2	4.7.6.2	1.5 dphu for the group
Insulation resistance	3.7.6.3	4.7.6.3	
Calibration adaptors	3.7.8	4.7.8	

* Defects per hundred units.

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection.

4.5.3.1.1 Three (3) samples of complete equipment shall be selected at random, one (1) sample for each Group 1, Group 2 and 4, and Group 3 in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. These samples shall constitute Group C requirement for first 100 units produced.

4.5.3.1.2 Thereafter, two (2) samples of complete equipment shall be selected at random, one (1) sample for each Group 2 and Group 3 in Table III from every 100 units and fraction thereof produced.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test (unless otherwise specified in the individual test) the sample shall be subject to and meet amplitude of oscillation per 3.7.2 and 4.7.2.

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4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government). The final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

TABLE III Group C inspection

Inspection	Req. Para.	Test Para.
Group 1		
Fungus	3.8.1	4.8.1
Group 2		
Low temperature	3.8.2	4.8.2
High temperature	3.8.3	4.8.3
Humidity	3.8.4	4.8.4
Group 3		
Altitude	3.8.5	4.8.5
Shock-bench handling	3.8.6	4.8.6
Vibration	3.8.7	4.8.7
Group 4		
Electromagnetic interference	3.8.8	4.8.8

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and C inspection may be accepted on contract or order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

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4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order provided units meet requirements of 4.5.4.

4.6. Bounce preconditioning. Each equipment, including accessories, produced shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.12.

4.7 Electrical.

4.7.1 Frequency range.

(a) The frequency band switch of the low frequency osciallator shall be set to the 10 to 20 position.

(b) The low frequency tuning dial shall be set to its full counter clockwise position.

(c) The proper resistor corresponding to the particular frequency band being used as indicated in 3.7.2 shall be inserted in the crystal socket.

(d) Set the screen voltage control to its maximum clockwise position.

(e) The frequency of oscillation shall be measured with a suitable electronic counter and shall meet the requirements of 3.7.1.

(f) Repeat (a) through (e) with the low frequency dial set to its mid-position.

(g) Repeat (a) through (e) with the low frequency dial set to its full clockwise position.

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(h) Repeat (a) through (g) with the low frequency band switch set to the "18 - 60" position.

(i) Set the high frequency band switch to the "55 - 75" position.

(j) Set the high frequency tuning dial to its full counter clockwise position.

(k) Repeat (c) through (e).

(l) Repeat (c) through (e) with the high frequency dial set to its mid-position.

(m) Repeat (c) through (e) with the high frequency dial set to its full clockwise position.

(n) Repeat (i) through (m) with the high frequency band switch to the "65 - 140" position.

4.7.2 Amplitude of oscillation.

(a) Insert the proper calibration resistor as indicated in 3.7.2 for Band 1.

(b) The grid meter shunt control is adjusted for a maximum grid current reading.

(c) When the tuning dial is rotated through the indicated frequency range for Band 1 the grid current reading shall meet the requirements of 3.7.2.

4.7.3 Resistance determination. A number of crystal units manufactured in accordance with MIL-C-3098 with frequencies in the range of 10 MHz to 140 MHz shall be measured for equivalent resistance in the Crystal Impedance Meter TS-683( )/TSM. The same crystals shall be measured for equivalent resistance in the Government furnished test model supplied to the contractor. The difference in resistance measurements shall meet the requirements of 3.7.3.

4.7.3.1 Drive level.

(a) Set the following controls in accordance with the specifications of the crystal to be tested. (Note 4.7.3.2(j)):

Frequency band switch and range switch to the nominal frequency of the crystal.
Meter shunt control fully clockwise.
Screen voltage control fully counter-clockwise.

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(b) Determine value of calibration resistance for the applicable crystal type from the specification covering the crystal to be tested. Select a resistor within ±5 percent of the specified value from the calibration resistors supplied with the TS-683( )/TSM. If the resistor is not available, mount a non-wire wound resistor of the required value with a tolerance of ±5 percent on the base of a HC-6/U crystal holder. This shall then take the place of the calibration resistance.

(c) Plug the resistor into the appropriate crystal socket. The crystal socket mounted to the left of the Grid current meter is used for crystals operating in the range of 10 MHz to 60 MHz. The one located to the right of the meter is used for crystals operating in the range of 55 MHz to 140 MHz.

(d) Turn the screen voltage control in a clockwise direction until a small grid current meter deflection is noted.

(e) Adjust the tuning to the nominal crystal frequency as indicated on electronic counter. Repeat (d) if oscillation ceases.

(f) Determine the required voltage drop across this calibration resistor by using the following formula:

$$E = \sqrt{PR_c}$$

Where:

E = voltage drop in volts.
P = specified drive level in watts.
R_c = calibration resistance in ohms.

(g) Insert the Adaptor, Crystal Socket MX-3020/TSM supplied with the TS-683( )/TSM into the appropriate crystal socket. Use the AC probe of an RF voltmeter to measure the voltage between each resistor pin and ground. Subtract the smaller reading from the larger to obtain the voltage drop "E" across the calibration resistor.

(h) If the measured voltage in (g) is the same as the computed voltage in (f), then the drive level is adjusted correctly.

(i) If the two voltages do not coincide, proceed as follows: Adjust the screen voltage control by turning it counterclockwise to decrease the measured voltage "E" and clockwise to increase the voltage "E" until the measured value of "E" coincides with the computed value of "E".

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(j) Do not change the setting of the screen voltage control after the drive level has been set.

(k) Remove calibration resistor.

4.7.3.2 Equivalent resistance of Series Resonant Crystals.

(a) Insert test crystal into the appropriate socket.

(b) Record grid current and frequency readings.

(c) Remove crystal and insert a fixed calibration resistor that will give a grid current reading as close as possible to that obtained in (b).

(d) Adjust tuning to the same frequency reading as in (b).

(e) Repeat (a) through (d) above until frequency readings are the same and grid current readings are as close as possible.

(f) Insert the crystal unit in its socket. Adjust the meter shunt control for a midscale meter reading. This current value shall be noted and designated I_{xtal}.

(g) Replace the crystal with a calibrating resistor that will give a grid current reading just below that obtained in (f). This current value shall be noted and designated I_l.

(h) Replace the calibrating resistor in (g) with one that will give a grid current reading just above that of (f). The current value shall be noted and designated I_h.

(i) The equivalent resistance of the crystal unit can then be calculated using the following formula:

$$R = R_l + \frac{(I_h - I_{xtal})(R_h - R_l)}{(I_h - I_l)}$$

Where:

R = equivalent resistance of the crystal unit.
R_l = resistance value of lower calibrating resistor.
R_h = resistance value of higher calibrating resistor.
I_{xtal} = grid current reading for crystal unit in microamperes.
I_l = grid current reading using R_h in microamperes.
I_h = grid current reading using R_l in microamperes.

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(j) Repeat 4.7.3.1 and 4.7.3.2 for the following crystal units:

Crystal	Freq. Range	Equiv. Resistance	Drive level
CR-52A/U	30 - 45 MHz	40 ohms	2.0 mw
CR-54A/U	50 - 87 MHz	60 ohms	2.0 mw
CR-54A/U	90 - 125 MHZ	50 - 60 ohms	2.0 mw

The resistance measured shall meet the requirements of 3.7.3.2.

(k) Insert the anti-resonant adaptor in the appropriate crystal crystal socket and plug the crystal unit into the adaptor.

(l) Repeat 4.7.3.1((a) thru (k)) and 4.7.3.2 ((a) thru (j)) for the following crystal:

Crystal	Freq. Range	Equiv. Resistance	Drive level
CR-18A/U	10 - 20 MHz	25 ohms	5 mw

Note: The adaptor along with the crystal must be removed when the anti-resonant resistance is to be determined. The resistance measured shall meet the requirements of 3.7.3.2.

4.7.4 Self-oscillation.

(a) Set the screen voltage control to its maximum clockwise position.

(b) Set the grid meter shunt control to its maximum clockwise position.

(c) Do not insert a crystal in the crystal socket.

(d) The crystal impedance meter shall be tuned throughout its entire frequency range.

(e) There shall be no indication of grid current different from residual grid current with the tuned circuits shunted out for rf, or presence of a signal as indicated by the means of a communications receiver.

4.7.5 Calibration resistors. The construction, values, and tolerances of the calibrating resistors shall be tested and meet the requirements of 3.7.5.

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4.7.6 RF pickup cord assembly.

4.7.6.1 Pull strength. After assembly of the cable, a pull shall be applied to the connector and shall meet the requirements of 3.7.6.1.

4.7.6.2 Dielectric strength. The test shall only be performed at the terminated end of the cable and shall meet the requirements of 3.7.6.2.

4.7.6.3 Insulation resistance. Using a suitable megohmeter the measured resistance shall meet the requirements of 3.7.6.3.

4.7.7 Power supply. Connect equipment to a power source capable of supplying ac voltages of 100 to 260 volts at frequencies of 50 to 400 Hz.

(a) Repeat 4.7.1 (and subparagraphs) with input voltage set to 103.5 volts, 115 volts, and 126.5 volts, respectively at a frequency of 50 Hz.

(b) Repeat (a) with the power source frequency set at 400 Hz.

(c) Repeat (a) and (b) with the power source set at 207, 230 and 253 volts respectively.

(d) The results shall meet the requirements of 3.7.7.

4.7.8 Calibration adaptors. The construction, values and tolerances, of the calibration adaptors shall be tested and shall meet the requirements of 3.7.8.

4.8 Service conditions. The Crystal Impedance Meter TS-683( )/TSM shall be tested in accordance with the following service conditions.

4.8.1 Fungus. The equipment, including accessories, in its normal operating configuration shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall meet the requirements of 3.8.1.

4.8.2 Low temperature. The equipment including accessories in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperatures in Step 2 held for 48 hours and in Step 4 held for 2 hours, and shall meet the requirements of 3.8.2.

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4.8.3 High temperature. The equipment including accessories in its normal operating configuration shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810, with temperature in Step 4 held for 2 hours, and shall meet the requirements of 3.8.3.

4.8.4 Humidity. The equipment including accessories in its normal operating configuration shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810, and shall meet the requirements of 3.8.4.

4.8.5 Altitude. The equipment including accessories in its normal operating configuration shall be subjected to the altitude test specified in Procedure I, Method 500 of MIL-STD-810, with altitude in Step 3 held for 1 hour, and shall meet the requirements of 3.8.5.

4.8.6 Shock, bench-handling. The equipment, less accessories and the case, shall be positioned as for servicing on a solid two inch fir bench top and shall be subjected to the following:

(a) Tilt up the equipment through an angle of 30 degrees, using one edge of the equipment as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis, for a total of four drops on horizontal face.

(b) Repeat (a) with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the equipment could be placed practicably during servicing.

(c) The equipment shall meet the requirements of 3.8.6.

4.8.7 Vibration. The equipment, including accessories, in its normal operating configuration shall be subjected to the vibration test as follows:

(a) Secure the equipment, less carrying case, and/or dust cover and/or transit case, directly to a vibration table than can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured.

(b) The equipment shall be vibrated with the frequency varying between 10 and 55 Hz at a total excursion of 0.03 ±0.006 inch, the frequency varying uniformly from 10 to 55 Hz and returning to 10 Hz in approximately 1 minute. Application of the vibration test specified above shall be in each of the following directions:

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(1) Horizontal, parallel to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(2) Horizontal, at right angles to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(3) Vertically to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(c) The equipment shall be inspected to ascertain that no fixed parts have become loose or damaged. The amplitude of oscillation measurements per 3.7.2 and 4.7.2 shall be performed after completion of vibration test, and the equipment shall meet the requirements of 3.8.7.

4.8.8 Electromagnetic interference. The equipment shall meet the requirements of 3.8.8. All test set-ups and procedures employed shall conform to the measurement techniques of MIL-STD-462.

4.9 Visual and mechanical. The complete equipment shall be examined for the defects listed in MIL-STD-252 and shall meet the requirements of 3.9.

4.10 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers and mountings.

(d) Location of crystal test holders, battery holder and terminals.

(e) Location of switches, sockets and fasteners.

(f) Location of cover and fasteners on carrying case which must mate.

(g) Location of panel mounting screw holes on basic unit for mating with case.

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4.11 Burn-in and debugging. Each equipment, less accessories, produced shall be placed in its normal operating position on a test table with room temperature at 77°F ±5°F, with specified operating voltage applied from a suitable power source. With all controls in the normal operating position, the equipment shall be turned on and remain on for a continuous 24 hour period. In case of parts failure, the equipment shall be repaired, recalibrated, and the burn-in and debugging test repeated and shall meet the requirements of 3.11. A failed item analysis report shall be submitted on each part failure with contents of report per 5.10.5 of MIL-STD-781.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Crystal Impedance Meter TS-683( )/TSM shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Crystal Impedance Meter TS-683( )/TSM shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Each Crystal Impedance Meter TS-683( )/TSM shall be packaged in accordance with items 1 through 16 of Table IV and Figure 1.

5.1.2 Level C. Each Crystal Impedance Meter TS-683( )/TSM shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2(b)).

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BILL OF MATERIAL TABLE

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Technical Literature	2	10-1	9-1/8	6								
	Bag	2		10-1/4	7		MIL-B-117	11	C				
2	Crystal Impedance Meter TS-683()/TSM	1	111	19	10-1/2	7	(Bare wt.19 Lbs.)						
3	Cell, Top	1		19	7	1-1/4	PPP-F-320	CF	WR	SW	W5c		
4	Lamp, Spare	1	111	1-3/16									
	Paperboard, Cushioning	1		3	3		PPP-P-291	1					
	Tape	1		4	1		PPP-T-76						
5	Lamp, Spare	1	111	1-3/16									
	Paperboard, Cushioning			3	3		PPP-P-291	1					
	Tape	1		4	1		PPP-T-76						
6	Fuse	6		1-1/4	1/4								
	Paperboard, Cushioning	1		4	4		PPP-P-291	1					
	Tape	1		4	1		PPP-T-76						
7	Tube, Electron	3	Package individually per MIL-B-75										
	Box, Fiber-board	1		4-1/4	2-7/8	1-1/2	PPP-B-636	CF	WR	SW	W6c		
			Consolidate three (3) packaged tubes within fiberboard box.										
	Tape	2		10	2		PPP-T-76						
8	Box containing:	1											
	Resistor	9	111										
	Adapter	4	111										
	Paperboard, Cushioning	2		7	6-1/4		PPP-P-291	1					
			Cushion Resistors together and adapters together. Secure with tape place within the metal box.										

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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CRYSTAL IMPEDANCE METER TS-683( )/TSM

MIL-C-10479B-(EL)

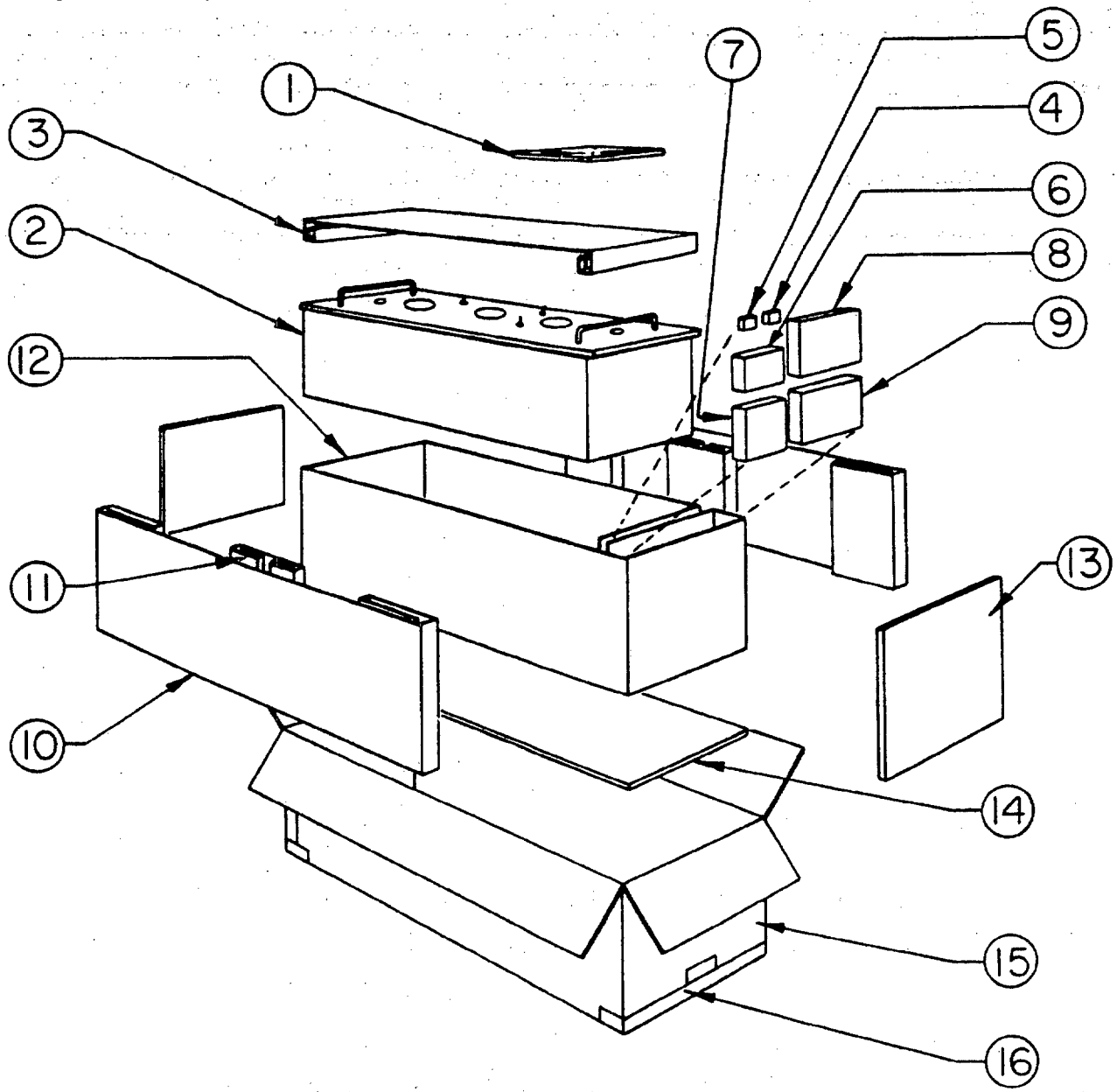


FIGURE 1

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5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of Crystal Impedance Meter TS-683( )/TSM, packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s. The load shall be"bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. Each Crystal Impedance Meter TS-683( )/TSM, packaged as specified in 5.1, shall be individually packed in a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, special requirements use, or a quantity shall be packed in a close-fitting fiberboard box conforming to MIL-B-43291 or PPP-B-640, class 2, style E. The gross weight of boxes conforming to PPP-B-640 shall not exceed 200 pounds. Closure shall be in accordance with the appendix to the applicable box specification. The shipping containers shall be reinforced by pressure-sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97, type IV, and PPP-S-760, type II respectively; selection of the material and application shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. A quantity of Crystal Impedance Meter TS-683( )/TSM packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C. Crystal Impedance Meter TS-683( )/TSM, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that the pallets, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Reinforcing of boxes, as specified in 5.2.1.2, will not be required. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The Crystal Impedance Meter TS-683( )/TSM is used to measure the equivalent electrical parameters of quartz crystals, within the range of 10 MHz to 140 MHz, of the type used for communication purposes.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Levels A, B or C) (see Section 5).

(c) If rough handling and functional tests are required.

(d) Quantity of first articles required.

(e) Place of final inspection.

(f) Technical literature required.

(g) Quantity of tools and running spare parts required.

(h) Marking for shipment and shipping containers.

6.3 Nomenclature and type designation. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design, for example: SG-12F/U or AN/URM-73C.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

6.5 Dimensional data. Dimensions of packaging materials in the Bill of Materials table are based on the dimensions of the equipment cited in the table. When dimensions of the equipment vary, the dimensions of the packaging materials shall be adjusted accordingly.

6.6 Temperature conversion. Temperature conversion from fahrenheit to centigrade as listed in this specification are as follows:

160°F	=	71°C
125°F	=	51.5°C
77°F	=	25°C
-4°F	=	-20°C
-65°F	=	-53.9°C

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6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodians:
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Preparing activity
Army - EL

Project Number 6625-A354

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-10479B(EL) CRYSTAL IMPEDANCE METER TS-683( )/TSM

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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